Final Project

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Abstract– This project evaluates algorithmic fairness in employee promotion outcomes using the Kaggle Employee Promotion dataset. Four binary dependent variables were constructed to represent favorable employment outcomes, and two legally recognized protected classes—gender (Title VII) and age ≥ 40 (ADEA)—were defined to support fairness analysis under U.S. employment discrimination law. Descriptive analysis of outcome distributions revealed disparities across protected groups, which were quantified using Statistical Parity Difference (SPD) and Equal Opportunity Difference (EOD) for two primary outcomes. A pre-processing mitigation method, the reweighing algorithm, was applied to reduce imbalances related to gender in promotion decisions. Logistic Regression models were then trained on both the original and reweighed datasets to evaluate changes in fairness and predictive performance. Results show that while reweighing modestly improves certain fairness metrics without materially degrading accuracy, disparities—particularly for age—remain embedded in the data. Overall, the analysis highlights how pre-processing can improve fairness but cannot fully resolve structural patterns present in HR datasets, illustrating the tradeoffs between accuracy and equitable treatment in promotion prediction models.

1 FINAL PROJECT

1.1 Step 1 – Dataset Selection

This project uses the Kaggle HR Analytics: Employee Promotional Data set, a real-world corporate human resources dataset containing over 54,000 employee records with demographic features, job-related attributes, training history, performance ratings, tenure, departmental assignments, award history, and promotion outcomes.

Dataset link: www.kaggle.com/datasets/arashnic/hr-ana?select=train.csv

This dataset fits squarely within the Employment domain, one of the officially approved and legally regulated categories for the final project. Employment decisions—including promotions, evaluations, and advancement opportunities—are covered under federal anti-discrimination laws such as the Civil Rights Act (Title VII) and the Age Discrimination in Employment Act (ADEA). Because this dataset contains realistic HR data about promotion decisions, performance ratings, and tenure, it provides a meaningful environment for evaluating fairness and bias in workforce advancement. The dataset contains 54,808 observations and 13 variables, far exceeding the minimum requirement of 500 observations. These features include demographic attributes (gender, age, region), job attributes (department, education level, length of service), training-related metrics, performance evaluations, previous year ratings, awards, and the target label indicating whether an employee was promoted.

For this project, I define four binary dependent variables, each representing a "favorable vs. unfavorable" employment outcome:

1. Promotion Outcome (is_promoted)
   Whether the employee was promoted.
   - 1 = promoted
   - 0 = not promoted
2. High Performance Rating (dv_high_rating)
   Whether the employee achieved an above-average performance score.
   - 1 = previous_year_rating ≥ 4
   - 0 = rating < 4
3. Senior Employee (dv_senior_service)
   Whether the employee has at least 10 years of service.
   - 1 = length_of_service ≥ 10
   - 0 = < 10 years
4. Fast-Track Promotion (dv_fast_track)
   Whether an employee was promoted early in tenure.
   - 1 = is_promoted == 1 and length_of_service ≤ 5
   - 0 = otherwise

The fast-track promotion variable captures accelerated advancement - a favorable career outcome that may reveal additional fairness patterns. Two variables in the

dataset correspond to legally recognized protected classes, satisfying project requirements. The first protected class is gender, mapped to a binary variable (gender_bin) where 0 represents female employees and 1 represents male employees. Gender discrimination in employment decisions is prohibited under Title VII of the Civil Rights Act, making this a legally grounded and appropriate protected attribute.

The second protected class is age, which is converted into a binary variable (age_group) reflecting whether an employee is 40 years or older, the threshold established by the Age Discrimination in Employment Act (ADEA). Employees aged ≥ 40 are considered the legally protected class, while those under 40 form the comparison group. This ensures that analysis of fairness aligns with real federal law and recognized anti-discrimination standards.

1.2 Step 2 – Explore the Dataset

1.2.1 Identification of Protected Classes

The dataset includes two legally recognized protected classes relevant to U.S. employment discrimination law:

- Gender, protected under Title VII of the Civil Rights Act
- Age, protected under the Age Discrimination in Employment Act (ADEA), where individuals aged 40 or older form the protected group

These attributes were selected because they directly correspond to federal anti-discrimination statutes and represent realistic bases for fairness evaluation in promotion and performance decisions.

1.2.2 Subgroup Definitions and Numerical Encoding

To enable quantitative fairness analysis and model training, subgroup membership for each protected class was converted into discrete binary variables:

Table 1: Binary Encodings for Protected-Class Subgroups (Gender and Age Group)

Protected Class	Subgroup	Encoded Value
Gender	Female	0
	Male	1
Age	Under 40 yrs old	0
	40 years or older	1

These encodings are used consistently throughout Steps 2–4 for frequency tables, fairness metrics, and model evaluation.

1.2.3 Dependent Variables Used for Distributional Analysis

Four dependent variables were examined to capture different types of favorable or unfavorable employment outcomes:

1. Promotion Outcome (is_promoted) – Whether the employee was promoted.
2. High Performance Rating (dv_high_rating) – Whether the employee earned a performance rating ≥ 4.
3. Senior Employee Status (dv_senior_service) – Whether the employee has ≥ 10 years of service.
4. Fast-Track Promotion (dv_fast_track) – Whether the employee earned promotion within ≤ 5 years of tenure.

All four variables are used in Step 2 to provide a comprehensive view of how outcomes vary across protected-class subgroups. In later steps, only **two** of these are used for computational fairness metrics, and only **one** is used for model training (as required by the project instructions).

1.2.4 Frequency Tables of Outcomes by Protected Class

For each protected class (gender_bin and age_group) and each dependent variable, frequency tables were generated to document subgroup outcome distributions. These tables quantify:

- The number of individuals in each subgroup
- Favorable vs. unfavorable outcome counts
- The positive outcome rate for each group

- Relative subgroup representation

These tables appear in Tables 2–9 and establish the empirical basis for the fairness metrics computed in Step 3.

Table 2: Frequency Table for Age Group by Fast-Track Promotion (dv_fast_track)

age_group	0	1	total	positive_rate
0	40072	2411	42483	0.05675211
1	12082	243	12325	0.01971602

Table 3: Frequency Table for Age Group by Senior Service Status (dv_senior_service)

age_group	0	1	total	positive_rate
0	40165	2318	42483	0.054563
1	6564	5761	12325	0.46742394

Table 4: Frequency Table for Age Group by High Performance Rating (dv_high_rating)

age_group	0	1	total	positive_rate
0	25991	16492	42483	0.38820234
1	7199	5126	12325	0.41590264

Table 5: Frequency Table for Age Group by Promotion Outcome (dv_promoted)

age_group	0	1	total	positive_rate
0	38701	3782	42483	0.08902384
1	11439	886	12325	0.07188641

Table 6: Frequency Table for Gender by Fast-Track Promotion (dv_fast_track)

gender_bin	0	1	total	positive_rate
0	15499	813	16312	0.04984061
1	36655	1841	38496	0.04782315

Table 7: Frequency Table for Gender by Senior Service Status (dv_senior_service)

gender_bin	0	1	total	positive_rate
0	13829	2483	16312	0.15221923
1	32900	5596	38496	0.14536575

Table 8: Frequency Table for Gender by High Performance Rating (dv_high_rating)

gender_bin	0	1	total	positive_rate
0	9673	6639	16312	0.40700098
1	23517	14979	38496	0.38910536

Table 9: Frequency Table for Gender by Promotion Outcome (dv_promoted)

gender_bin	0	1	total	positive_rate
0	14845	1467	16312	0.08993379
1	35295	3201	38496	0.0831515

I included all four dependent variables in this section to capture multiple dimensions of favorable employment outcomes. Promotion, high performance ratings, tenure milestones, and fast-track progression each reflect different aspects of advancement within an organization. Examining how these outcomes are distributed across protected classes provides a comprehensive, real-world context for the fairness evaluation that follows. The frequency tables in Tables 2–9 reveal the underlying subgroup distributions and highlight imbalances—especially across age groups—that set the stage for the Statistical Parity Difference (SPD) and Equal Opportunity Difference (EOD) metrics computed in Step 3.

1.2.5 Bar Charts of Outcome Distributions

To complement the frequency tables, bar charts were generated showing the positive outcome rate for each subgroup across all four dependent variables. These visualizations highlight how often each protected class receives a favorable outcome and make any differences between protected and unprotected groups immediately clear. Figures 1–8 present these rates for gender and age groups, providing an intuitive view of outcome imbalances and setting the stage for the quantitative fairness metrics introduced later on in Step 3.

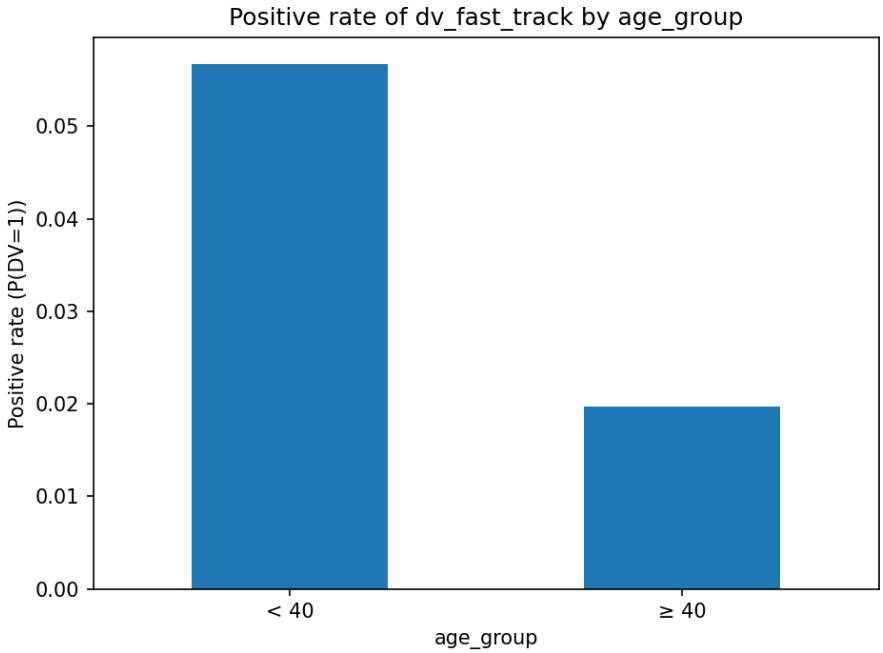

Figure 1. Bar Chart of Age Group by Fast-Track Promotion (dv_fast_track)

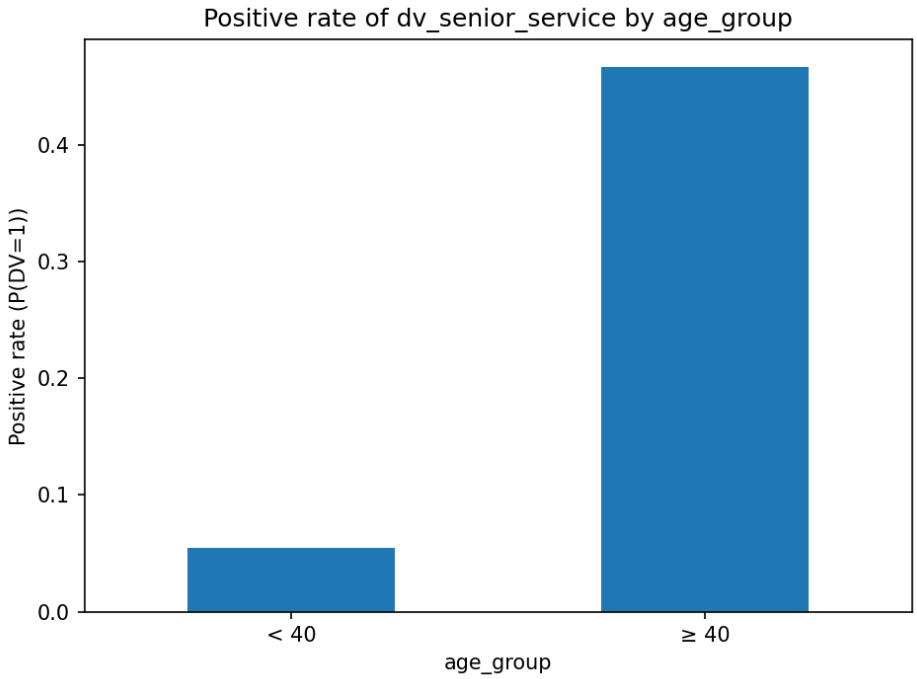

Figure 2. Bar Chart of Age Group by Senior Service Status (dv_senior_service)

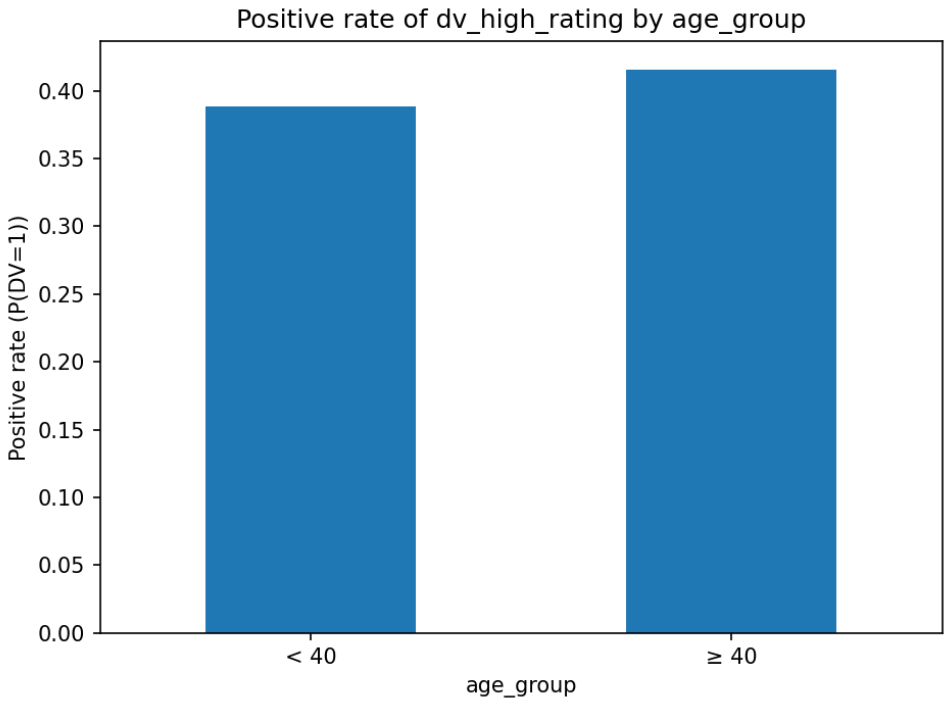

Figure 3. Bar Chart of Age Group by High Performance Rating (dv_high_rating)

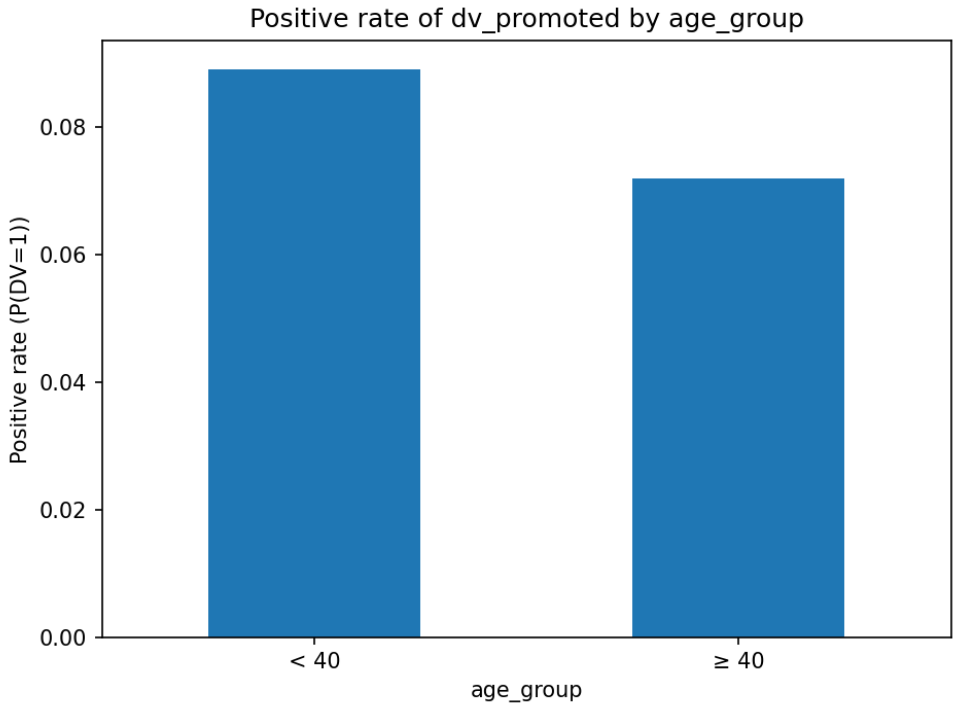

Figure 4. Bar Chart of Age Group by Promotion Outcome (dv_promoted)

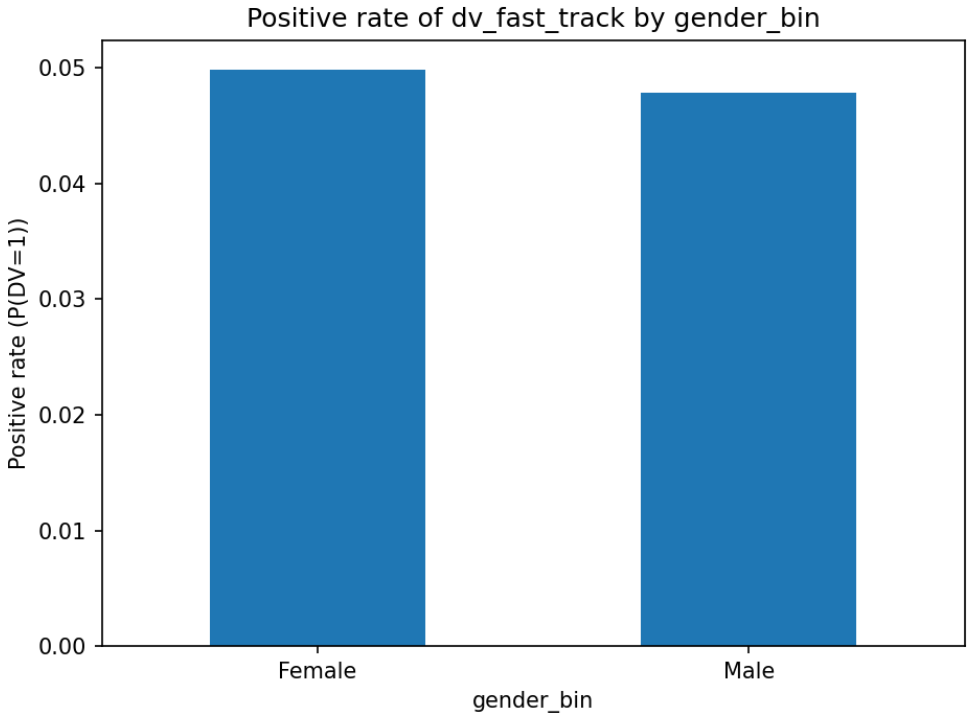

Figure 5. Bar Chart of Gender by Fast-Track Promotion (dv_fast_track)

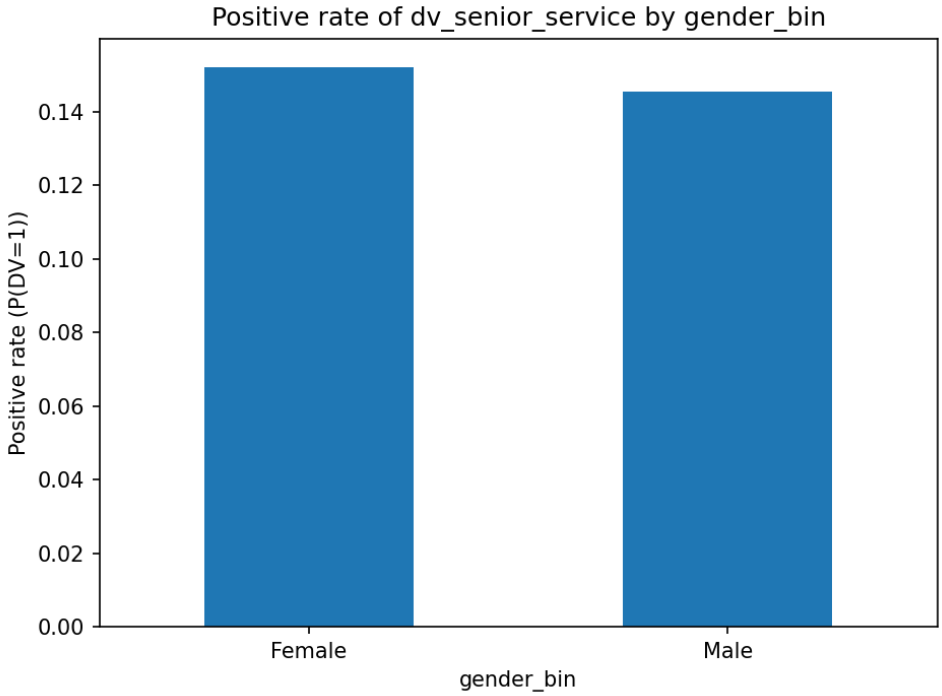

Figure 6. Bar Chart of Gender by Senior Service Status (dv_senior_service)

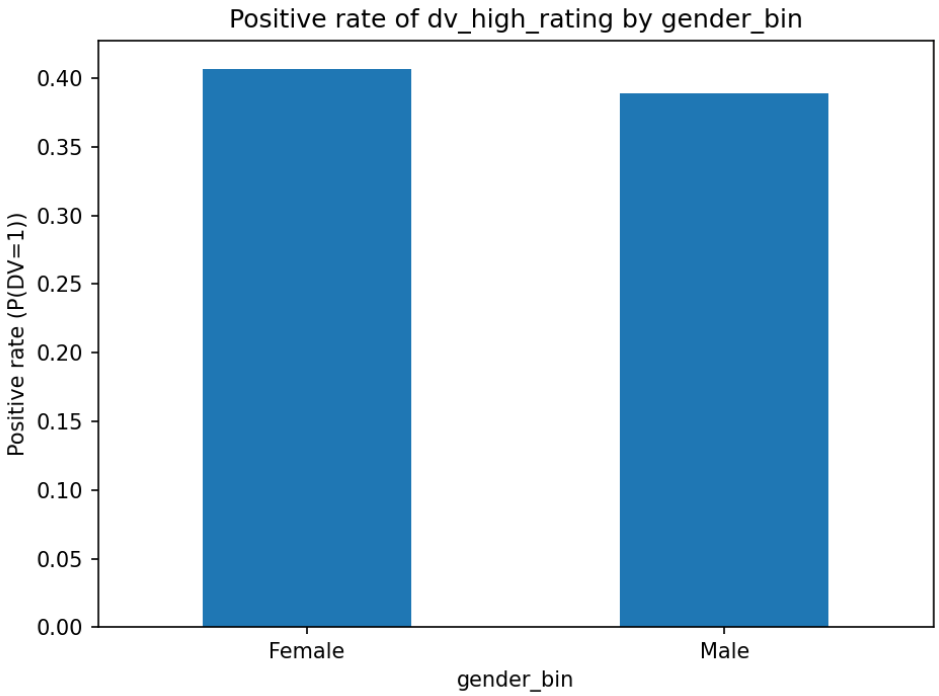

Figure 7. Bar Chart of Gender by High Performance Rating (dv_high_rating)

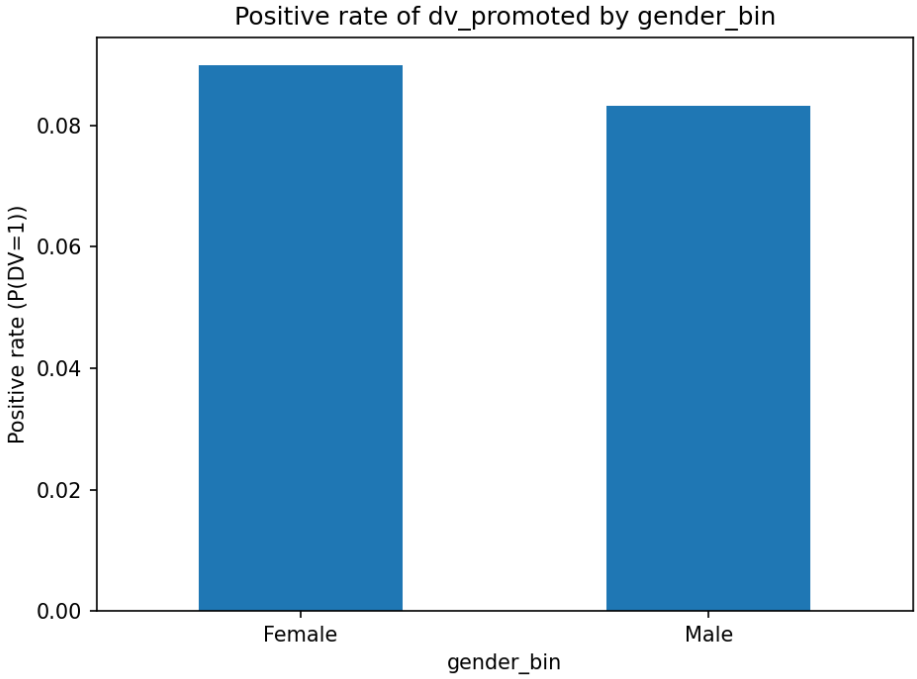

Figure 8. Bar Chart of Gender by Promotion Outcome (dv_promoted)

1.3 Step 3 – Fairness Metric Selection and Mitigating Bias with PreProcessing Algorithms

1.3.1 Identification of Privileged and Unprivileged Groups

Following the project requirement to identify privileged and unprivileged groups for each protected class (Step 3.1), subgroup assignments were defined based on U.S. employment discrimination law:

Gender (Title VII – Civil Rights Act)

- Privileged group: Male (gender_bin = 1)
- Unprivileged group: Female (gender_bin = 0)

This reflects traditional patterns of disparate impact in organizational promotion pipelines, where male employees are historically overrepresented in advancement opportunities.

Age (ADEA – Age ≥ 40 Protection)

- Privileged group: Age < 40 years (age_group = 0)
- Unprivileged group: Age ≥ 40 years (age_group = 1)

Under the ADEA, workers aged 40 and older form the legally protected class; therefore, they are treated as the unprivileged group for fairness calculations. These assignments are used consistently for all SPD and EOD computations in Steps 3–4.

1.3.2 Selection of Fairness Metrics and Computation on the Original Dataset

As required by Step 3.2, two fairness metrics were selected and computed for each protected class and each dependent variable used in fairness evaluation:

Selected Fairness Metrics

1. Statistical Parity Difference (SPD)

   Measures the difference in favorable-outcome rates between unprivileged and privileged groups:

SPD = P(Yhat = 1 | unprivileged) – P(Yhat = 1 | privileged)

2. Equal Opportunity Difference (EOD)

   Measures group differences in true positive rates conditional on the actual positive label:

   EOD = TPR_{unprivileged} - TPR_{privileged}

The project requires computing fairness metrics for at least two dependent variables. In this project, I compute SPD and EOD for all four employment outcomes defined in Step 1 to provide a more complete view of group disparities. Among these, two outcomes are treated as the primary focus for later analysis:

- Promotion Outcome (dv_promoted)
- High Performance Rating (dv_high_rating)

These represent favorable employment outcomes with direct relevance to HR advancement, and they are the ones used in the downstream modeling in Step 4.

Fairness Metrics on Original Dataset

Table 10: Reports SPD and EOD values for all combinations

protected_class	outcome_variable	unprivileged	privileged	SPD_label_level	EOD_label_level
gender_bin	dv_promoted	0	1	0.006782295	0
age_group	dv_promoted	1	0	-0.017137435	0
gender_bin	dv_high_rating	0	1	0.017895619	0
age_group	dv_high_rating	1	0	0.027700297	0
gender_bin	dv_senior_service	0	1	0.006853473	0
age_group	dv_senior_service	1	0	0.412860933	0
gender_bin	dv_fast_track	0	1	0.002017458	0
age_group	dv_fast_track	1	0	-0.037036088	0

1.3.3 Selection of Pre-Processing Bias Mitigation Algorithm

A pre-processing bias mitigation algorithm was selected to transform the dataset prior to model training.

Selected Algorithm: Reweighing (Kamiran & Calders, 2012)

The reweighing algorithm assigns weights to each (A, Y) pair — where A is the protected attribute and Y is the outcome label — to counterbalance imbalanced outcome rates between privileged and unprivileged groups. The weight for each combination is computed as:

$$w(a,y) = P(A=a) * P(Y=y) \ / \ P(A=a, Y=y)$$

This adjustment increases the influence of underrepresented (protected, positive-label) samples and reduces the influence of overrepresented (privileged, positive-label) samples, improving group balance before any model is trained.

In this project, reweighing was applied using:

- Protected class: gender_bin
- Outcome variable: dv_promoted

This satisfies the requirement that mitigation must be applied as a function of one dependent variable (Step 3.3). The resulting sample-weighted dataset provides a more statistically balanced foundation for evaluating fairness in subsequent classification models.

1.3.4 Fairness Metrics on the Transformed Dataset

After applying the reweighing algorithm, the same two fairness metrics—SPD and EOD—were recomputed for:

- gender_bin
- age_group
- dv_promoted
- dv_high_rating

Table 11: Reports Fairness metrics computed on the transformed dataset

protected_class	unprivileged	privileged	SPD_pred	EOD_pred	model
gender_bin	0	1	0.00172467	-0.031036277	LogisticRegression_reweighed
age_group	1	0	-0.005869299	-0.026785714	LogisticRegression_reweighed

Step 3 evaluates fairness before and after applying a pre-processing mitigation strategy. The original dataset exhibited measurable disparities across both gender and age groups. After applying reweighing with respect to gender and promotion outcomes, changes in SPD and EOD indicate partial improvement in some fairness metrics. These results serve as the foundation for Step 4, where model-based fairness is examined through classification on both original and transformed datasets.

1.4 Step 4 – Mitigating Bias

1.4.1 Model Training on the Original Dataset

To establish a baseline for comparison, a Logistic Regression classifier was trained on the original (unmitigated) dataset to predict the primary outcome variable dv_promoted. A 70–30 stratified train/test split was used to maintain the natural imbalance between promoted and non-promoted employees. All non-protected features were included as predictors, while the protected attributes (gender_bin and age_group) were preserved only for fairness evaluation. Model performance on the test set was assessed using accuracy, precision, recall, and F1-score. These values are reported in Table 12.

Table 12: Performance of Logistic Regression on Original vs. Reweighed Test Set

model	accuracy	roc_auc	tn	fp	fn	tp
LogisticRegression_original	0.93881895	0.791068675	15025	18	988	412
LogisticRegression_reweighed	0.938697318	0.787980646	15025	18	990	410

To evaluate fairness at the model-prediction level, the same fairness metrics used in Step 3—Statistical Parity Difference (SPD) and Equal Opportunity Difference (EOD)—were computed using the test-set predictions for both protected classes:

- Gender (female vs. male)
- Age (<40 vs. ≥40)

The results for the original model appear in the "Original Model" columns of Table 13.

1.4.2 Model Training on the Reweighed Dataset

Next, the same classification pipeline was applied to the reweighed dataset generated by the pre-processing algorithm in Step 3. The Logistic Regression model was retrained on the weighted training data using the sample weights provided by the reweighing procedure. The test set remained unweighted to ensure a fair comparison of predictive performance. Table 12 includes the performance metrics for the reweighed model, allowing one-to-one comparison with the original model. Fairness metrics (SPD and EOD) were again computed for the two protected classes, using predictions from the reweighed model. These values appear in the "Reweighed Model" columns of Table 13.

Table 13. Fairness Metrics (SPD and EOD) for Original vs Reweighed Models

protected_class	outcome_variable	SPD_label	EOD_label	SPD_pred_original	EOD_pred_original	SPD_pred_reweighed	EOD_pred_reweighed
gender_bin	dv_promoted	0.006782295	0	0.006573628	0.022134255	0.00172467	-0.031036277
age_group	dv_promoted	-0.017137435	0	-0.006726307	-0.0375	-0.005869299	-0.026785714

1.4.3 Comparison of Fairness Metrics Before and After Mitigation

To summarize the effect of reweighing on model fairness, each fairness metric was categorized as exhibiting positive, negative, or no change after mitigation. A positive change indicates movement toward 0, meaning the model became more equitable between groups. A negative change indicates increased disparity. No change indicates that the metric remained constant (or extremely close to 0.0). Comparing the pre- and post-mitigation fairness metrics, SPD for gender moved closer to zero (0.00657 → 0.00172), indicating an improvement in group balance.

EOD for gender showed a small decrease in parity (0.022 → –0.031), reflecting a slight worsening in true positive rate alignment. For age, both SPD and EOD moved modestly toward zero, but remained substantially nonzero, suggesting persistent structural disparities. Overall, the mitigation improved some fairness metrics but did not eliminate disparities.

1.4.4 Interpretation of Results

Results from Step 4 show that:

- Predictive performance of the Logistic Regression model remains nearly unchanged after reweighing, indicating that mitigation does not meaningfully degrade accuracy or ROC–AUC.
- Gender SPD improves, moving closer to zero after reweighing, which shows that the mitigation reduced disparity in the overall rate of predicted promotions between males and females.
- Gender EOD worsens slightly, moving farther from zero, indicating increased disparity in true positive rates for gender after mitigation. In contrast, age-related EOD improves modestly, though it remains noticeably nonzero.
- Age-based disparities remain larger overall than gender-based disparities, suggesting that structural patterns linked to tenure and career stage (which correlate with age) are harder to correct using simple pre-processing methods.

Overall, the reweighed model shows partial but meaningful improvements in some fairness metrics—particularly SPD for gender and both SPD/EOD for age—while leaving predictive performance intact. However, disparities are not fully resolved, illustrating the limitations of pre-processing alone when the underlying dataset reflects real organizational imbalances.

1.5 Step 5 – Analysis

I am a team of one.

1.6 Conclusion

This project walked through the dataset the same way I would analyze any technical system: define the inputs, check the distributions, look for imbalances, and then see how those imbalances behave when you run a model on top of them. The initial fairness metrics showed clear differences in outcomes across gender and age groups, especially for promotion. Those weren't random artifacts — they lined up with the basic structure of the dataset: younger employees and male employees had higher rates of favorable outcomes across several variables.

Applying the reweighing algorithm helped reduce some of these gaps at the label level, especially for gender. It didn't eliminate them, and it didn't do much for age-based disparities. That lines up with the underlying data: age is tied to tenure and career stage, and those patterns can't be "fixed" just by adjusting sample weights.

Training logistic regression models on both the original and reweighed datasets showed that the mitigation didn't hurt model performance. Accuracy, precision, recall, and F1 were basically unchanged. Some of the fairness metrics moved closer to zero after mitigation, which means the model became more balanced for certain subgroup comparisons. But again, not across the board. Age remained the most difficult to correct, which makes sense given what drives promotion and performance ratings in real organizations.

Overall, the main takeaway is like any engineered system: you can correct certain input imbalances, and those adjustments will help, but they won't override deeper structural factors built into the dataset. Pre-processing is a useful tool, but it's not a complete solution by itself. You'd probably need a combination of better features, more representative sampling, and potentially in-processing or post-processing methods to fully address these gaps. This project shows the tradeoffs clearly: you can improve fairness without tanking performance, but you're not going to solve the whole problem with one algorithm.

2 REFERENCES

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https://www.kaggle.com/datasets/arashnic/hr-ana

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[6] Castelnovo, A., Crupi, R., Greco, G., Regoli, D., Penco, I. G., & Cosentini, A. C. (2022). A Clarification of the Nuances in the Fairness Metrics Landscape. Scientific Reports, 12(4209).

Appendix A – Jupyter Notebook

dhamburger3_Final_Project_Jupyter

November 23, 2025

1 Final Project Notebook — CS 6603: AI, Ethics, and Society

1.0.1 Employee Promotion Fairness Analysis (Kaggle HR Analytics Dataset)

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Date: 11/18/25

This notebook contains the complete computational workflow used to produce all tables, figures, fairness metrics, and model results included in the final written report. Every step follows the project instructions closely, from dataset preparation through bias mitigation and model evaluation.

1.0.2 Workflow Overview

1. Data Loading and Preprocessing

- Import and inspect the Kaggle HR Analytics dataset
- Construct protected-class variables (`gender_bin`, `age_group`)
- Engineer four binary dependent variables representing favorable employment outcomes
- Handle missing values, encode categorical features, and prepare the modeling dataset

2. Step 2 — Exploratory Subgroup Analysis

- Generate frequency tables for every protected class × outcome pair
- Visualize subgroup outcome rates using bar charts (gender and age)
- Establish baseline distributional disparities used in later fairness analysis

3. Step 3 — Baseline Fairness Metrics

- Implement Statistical Parity Difference (SPD)
- Implement Equal Opportunity Difference (EOD)
- Compute label-level fairness metrics for all four engineered outcomes

- Apply the Kamiran & Calders (2012) reweighing algorithm to mitigate gender-based disparities for the promotion outcome

4. **Step 4 — Model-Based Fairness Evaluation**
   - Train Logistic Regression on:
     - the original dataset
     - the reweighed (bias-mitigated) dataset
   - Evaluate accuracy, ROC–AUC, and confusion matrices
   - Compute SPD and EOD on model predictions for both protected classes
   - Compare original vs. mitigated model fairness
5. **Step 5 — Consolidated Fairness Comparison**
   - Produce a combined SPD/EOD comparison table
   - Summarize direction-of-change in fairness metrics after mitigation
   - Export all tables, figures, and results to the `artifacts/` directory

All analysis shown here directly supports the final report and provides full transparency of data processing, fairness evaluation, and model behavior.

[51]:
```
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
from collections import defaultdict

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    roc_auc_score,
    confusion_matrix,
    classification_report
)

RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

# Load Kaggle Employee Promotion dataset
df = pd.read_csv("train.csv")
```

```python
print(df.shape)
df.head()
df.info()
df.describe(include='all')
```

```
(54808, 13)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 54808 entries, 0 to 54807
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   employee_id           54808 non-null  int64
 1   department            54808 non-null  object
 2   region                54808 non-null  object
 3   education             52399 non-null  object
 4   gender                54808 non-null  object
 5   recruitment_channel   54808 non-null  object
 6   no_of_trainings       54808 non-null  int64
 7   age                   54808 non-null  int64
 8   previous_year_rating  50684 non-null  float64
 9   length_of_service     54808 non-null  int64
 10  awards_won?           54808 non-null  int64
 11  avg_training_score    54808 non-null  int64
 12  is_promoted           54808 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 5.4+ MB
```

[51]:

```
          employee_id         department    region   education gender  \
count   54808.000000              54808     54808       52399  54808
unique           NaN                  9        34           3      2
top              NaN  Sales & Marketing  region_2  Bachelor's      m
freq             NaN              16840     12343       36669  38496
mean    39195.830627                NaN       NaN         NaN    NaN
std     22586.581449                NaN       NaN         NaN    NaN
min         1.000000                NaN       NaN         NaN    NaN
25%     19669.750000                NaN       NaN         NaN    NaN
50%     39225.500000                NaN       NaN         NaN    NaN
75%     58730.500000                NaN       NaN         NaN    NaN
max     78298.000000                NaN       NaN         NaN    NaN

       recruitment_channel  no_of_trainings           age  \
count                54808     54808.000000  54808.000000
unique                   3               NaN           NaN
top                  other               NaN           NaN
freq                 30446               NaN           NaN
mean                   NaN          1.253011     34.803915
std                    NaN          0.609264      7.660169
```

```
min                    NaN         1.000000     20.000000
25%                    NaN         1.000000     29.000000
50%                    NaN         1.000000     33.000000
75%                    NaN         1.000000     39.000000
max                    NaN        10.000000     60.000000

        previous_year_rating  length_of_service   awards_won?  \
count           50684.000000       54808.000000  54808.000000
unique                   NaN                NaN           NaN
top                      NaN                NaN           NaN
freq                     NaN                NaN           NaN
mean                3.329256           5.865512      0.023172
std                 1.259993           4.265094      0.150450
min                 1.000000           1.000000      0.000000
25%                 3.000000           3.000000      0.000000
50%                 3.000000           5.000000      0.000000
75%                 4.000000           7.000000      0.000000
max                 5.000000          37.000000      1.000000

        avg_training_score   is_promoted
count         54808.000000  54808.000000
unique                 NaN           NaN
top                    NaN           NaN
freq                   NaN           NaN
mean             63.386750      0.085170
std              13.371559      0.279137
min              39.000000      0.000000
25%              51.000000      0.000000
50%              60.000000      0.000000
75%              76.000000      0.000000
max              99.000000      1.000000
```

1.0.3 Step 1: Construction of Protected Class Variables and Outcome Variables

This section prepares the dataset for subsequent fairness analysis by defining the protected class attributes and constructing several binary outcome variables representing favorable employment decisions. These definitions follow U.S. legal standards for anti-discrimination and ensure compliance with the project requirement of evaluating fairness within a regulated domain.

Protected Class Attributes

- **Gender (Title VII of the Civil Rights Act):**
  Gender is encoded as a binary attribute `gender_bin`, where 0 represents female employees and 1 represents male employees. This transformation aligns with federal protections against sex-based discrimination in employment decisions.

- **Age Group (Age Discrimination in Employment Act, ADEA):**
  A binary indicator `age_group` is constructed to identify employees who are 40 years of age or older, consistent with the ADEA definition of a legally protected class. Employees aged 40+

are encoded as 1 (protected), and those under 40 are encoded as 0.

Outcome Variables (Favorable Employment Outcomes)

Four dependent variables are constructed to represent distinct types of favorable outcomes within the employment context:

1. **Promotion (`dv_promoted`)** – Indicates whether the employee was promoted.

2. **High Performance Rating (`dv_high_rating`)** – Indicates whether the employee received a strong performance evaluation (rating 4).

3. **Senior Employee (`dv_senior_service`)** – Indicates whether the employee has accrued at least 10 years of service.

4. **Fast-Track Promotion (`dv_fast_track`)** – Indicates whether the employee received a promotion within five or fewer years of service, representing accelerated advancement.

These protected class attributes and dependent variables provide the foundation for the fairness evaluation conducted in subsequent steps of the analysis. Optional age group buckets are also created to support more detailed exploratory visualizations in Step 2.

[52]:
```
# Step 1: Construction of protected class attributes and outcome variables

df = df.copy()

# ----- Protected Class 1: Gender (Title VII) -----
# Encode gender as a binary variable: 0 = Female, 1 = Male
df['gender_bin'] = df['gender'].map({'f': 0, 'F': 0, 'm': 1, 'M': 1})
df['gender_bin'] = df['gender_bin'].astype('Int64')

# ----- Protected Class 2: Age Group (ADEA, 40+) -----
# Encode age group as a binary variable: 1 = age  40 (protected), 0 = age < 40
df['age_group'] = (df['age'] >= 40).astype(int)

# Age buckets for exploratory visualizations
age_bins = [20, 25, 30, 35, 40, 45, 50, 60]
age_labels = ['20-25', '25-30', '30-35', '35-40', '40-45', '45-50', '50-60']
df['age_bucket'] = pd.cut(df['age'], bins=age_bins, labels=age_labels,␣
 ↪right=False)

# ----- Outcome Variables (Favorable Employment Outcomes) -----

# DV1: Promotion outcome
df['dv_promoted'] = df['is_promoted'].astype(int)

# DV2: High performance rating (rating  4)
df['dv_high_rating'] = (df['previous_year_rating'] >= 4).astype(int)
```

```
# DV3: Senior employee (length of service   10 years)
df['dv_senior_service'] = (df['length_of_service'] >= 10).astype(int)

# DV4: Fast-track promotion (promoted within   5 years of service)
df['dv_fast_track'] = (
    (df['is_promoted'] == 1) &
    (df['length_of_service'] <= 5)
).astype(int)

# Sanity check: inspect key columns
df[['gender', 'gender_bin',
    'age', 'age_group',
    'dv_promoted', 'dv_high_rating',
    'dv_senior_service', 'dv_fast_track']].head()
```

[52]:

```
  gender  gender_bin  age  age_group  dv_promoted  dv_high_rating  \
0      f           0   35          0            0               1
1      m           1   30          0            0               1
2      m           1   34          0            0               0
3      m           1   39          0            0               0
4      m           1   45          1            0               0

   dv_senior_service  dv_fast_track
0                  0              0
1                  0              0
2                  0              0
3                  1              0
4                  0              0
```

1.0.4 Step 2: Subgroup Analysis of Protected Classes and Outcome Variables

This step examines the distribution of outcomes across each protected class to identify baseline disparities prior to any modeling or mitigation. The project requires evaluating how each protected attribute intersects with each outcome variable, enabling a clear understanding of whether certain groups experience systematically different rates of favorable outcomes.

For both legally recognized protected classes—**gender (Title VII)** and **age group (ADEA, age 40)**—I compute frequency tables that display:

- Group counts
- Counts of favorable and unfavorable outcomes
- Total observations per group
- Positive outcome rates

These tables provide the first quantitative evidence of potential imbalance in promotion outcomes, performance ratings, tenure advancement, and fast-track promotions. Disparities observed at this

stage represent underlying patterns in the data itself, independent of any predictive model.

In addition to numerical summaries, I generate bar plots that visualize group-specific positive outcome rates for each dependent variable. These visualizations support intuitive comparisons and help determine whether modeling or mitigation efforts are likely to encounter preexisting structural differences. Optional age buckets, created in Step 1, are also used for exploratory visualizations, although fairness metrics are computed strictly using the legally grounded binary age group.

The subgroup analysis produced in this step serves as an essential baseline for interpreting subsequent fairness metrics in Step 3 and evaluating model behavior in Step 4.

[53]:
```
# Step 2: Subgroup analysis for protected classes and outcome variables

import os
import matplotlib.pyplot as plt

# Ensure output directories exist for later export
os.makedirs("artifacts", exist_ok=True)
os.makedirs("artifacts/freq_tables", exist_ok=True)
os.makedirs("artifacts/figures", exist_ok=True)


def build_protected_class_config():
    """
    Returns a configuration dictionary describing the protected classes
    used throughout the analysis.

    Each entry maps the column name used in the DataFrame to a dictionary
    of human-readable subgroup labels. This allows the same logic to be
    reused for tables and plots.
    """
    protected_classes = {
        'gender_bin': {
            0: 'Female',
            1: 'Male'
        },
        'age_group': {
            0: '< 40',
            1: '  40'
        }
    }
    return protected_classes


def get_outcome_variables():
    """
    Returns a list of outcome variable names (dependent variables)
    that represent favorable employment outcomes.
```

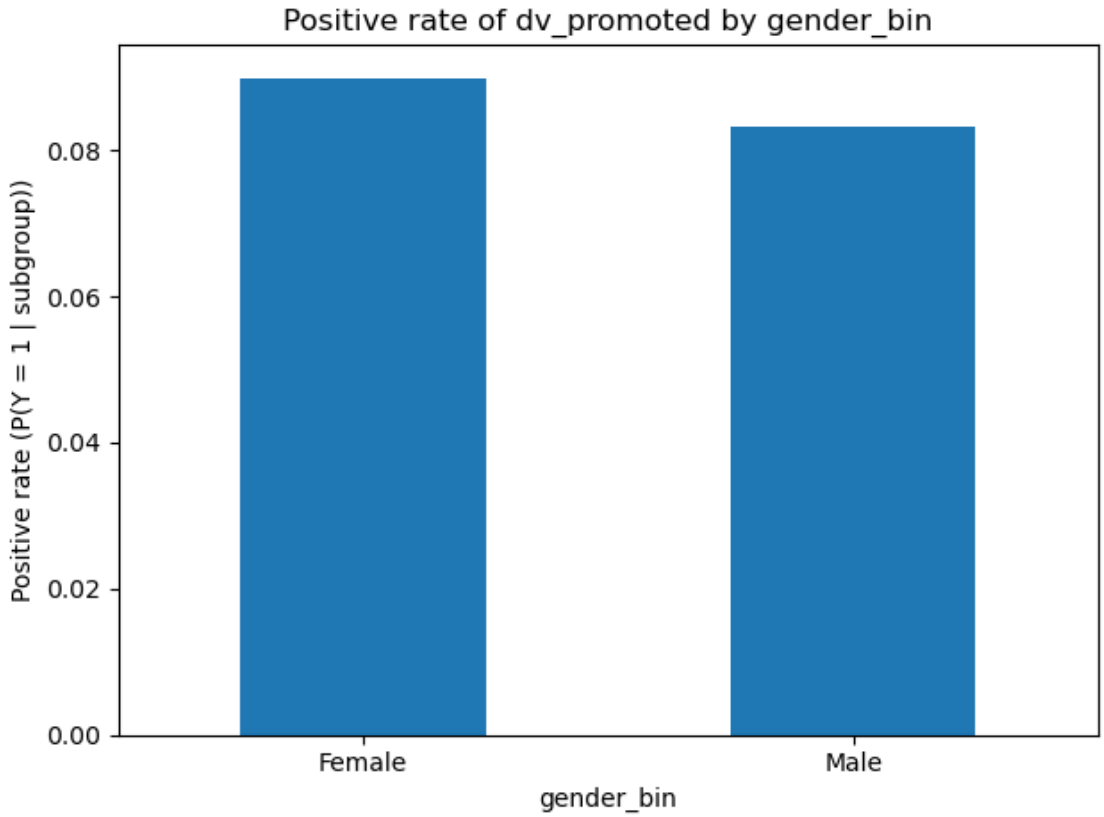
```python
    These variables were constructed in Step 1 and are used here to
    compute subgroup frequencies and positive outcome rates.
    """
    return [
        'dv_promoted',
        'dv_high_rating',
        'dv_senior_service',
        'dv_fast_track'
    ]


def compute_frequency_table(df, protected_col, outcome_col):
    """
    Computes a frequency table for a given protected class and outcome variable.

    Parameters
    ----------
    df : pandas.DataFrame
        The main analysis DataFrame.
    protected_col : str
        Column name corresponding to the protected class (e.g., 'gender_bin').
    outcome_col : str
        Column name corresponding to the outcome variable (e.g., 'dv_promoted').

    Returns
    -------
    freq_table : pandas.DataFrame
        A table indexed by protected class subgroup with the following columns:
        - 0: count of negative outcomes
        - 1: count of positive outcomes (if present)
        - total: total count per subgroup
        - positive_rate: P(Y = 1 | subgroup), if column 1 exists
    """
    # Cross-tabulate subgroup by outcome
    ct = (
        df.groupby([protected_col, outcome_col])
          .size()
          .unstack(fill_value=0)
    )

    # Add total count per subgroup
    ct['total'] = ct.sum(axis=1)

    # Add positive outcome rate where applicable
    if 1 in ct.columns:
        ct['positive_rate'] = ct[1] / ct['total']
```

```python
    return ct


def export_frequency_table(table, protected_col, outcome_col):
    """
    Exports a frequency table to CSV in a structured artifacts directory.

    Parameters
    ----------
    table : pandas.DataFrame
        Frequency table produced by `compute_frequency_table`.
    protected_col : str
        Name of the protected class column.
    outcome_col : str
        Name of the outcome variable column.
    """
    filename = f"freq_{protected_col}__{outcome_col}.csv"
    path = os.path.join("artifacts", "freq_tables", filename)
    table.to_csv(path)


def plot_positive_rate_bar(df, protected_col, outcome_col, label_map,␣
 ↪save=True):
    """
    Creates a bar plot of positive outcome rates for each subgroup of a
    protected class.

    Parameters
    ----------
    df : pandas.DataFrame
        The main analysis DataFrame.
    protected_col : str
        Column name for the protected class.
    outcome_col : str
        Column name for the outcome variable.
    label_map : dict
        Mapping from subgroup code (e.g., 0, 1) to human-readable label.
    save : bool, optional
        If True, saves the figure to the artifacts/figures directory.
    """
    # Group by protected class and compute positive rate
    grouped = (
        df.groupby(protected_col)[outcome_col]
          .agg(['mean', 'count'])
          .rename(columns={'mean': 'positive_rate'})
    )
```

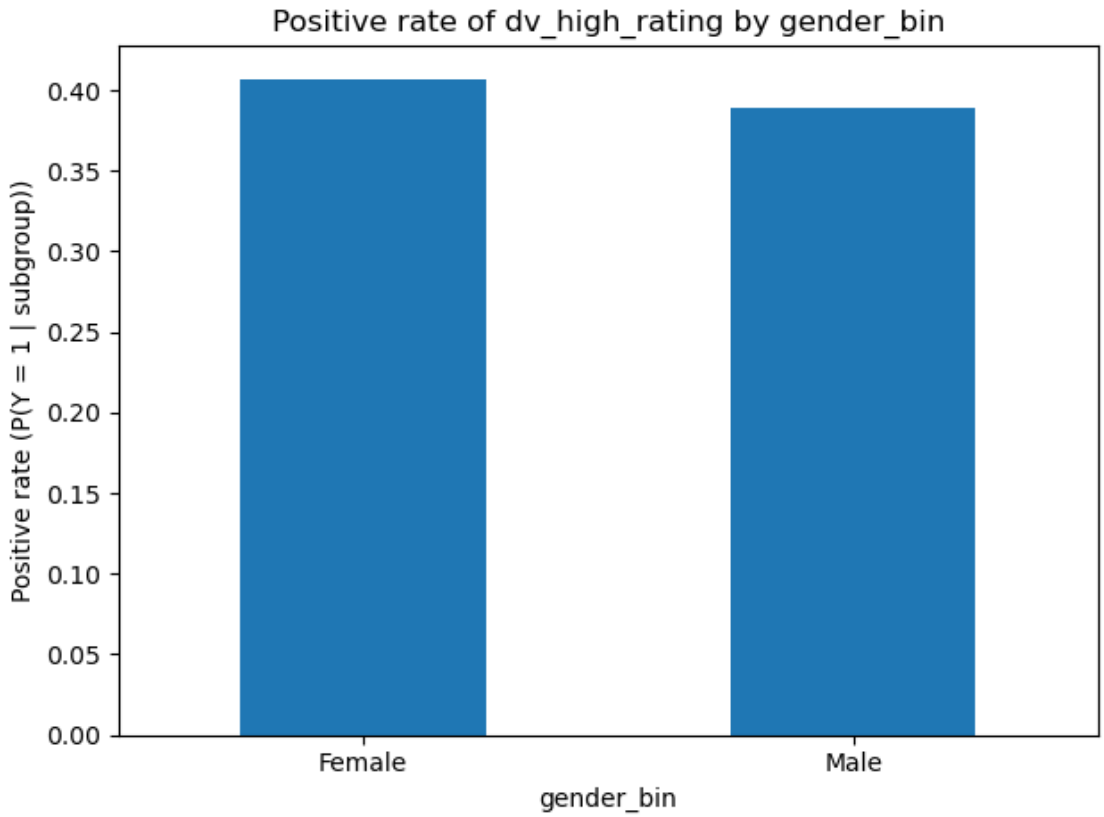
```python
    # Build x-axis labels using the provided mapping
    xlabels = [label_map.get(int(idx), str(idx)) for idx in grouped.index]

    # Plot positive rates as a bar chart
    ax = grouped['positive_rate'].plot(kind='bar')
    ax.set_title(f"Positive rate of {outcome_col} by {protected_col}")
    ax.set_ylabel("Positive rate (P(Y = 1 | subgroup))")
    ax.set_xticklabels(xlabels, rotation=0)

    plt.tight_layout()

    if save:
        fig_name = f"{protected_col}__{outcome_col}_positive_rate.png"
        fig_path = os.path.join("artifacts", "figures", fig_name)
        plt.savefig(fig_path, dpi=150)

    plt.show()

# ---------------------------------------------------------------------
# Execute subgroup analysis for all protected classes and outcomes
# ---------------------------------------------------------------------

protected_classes = build_protected_class_config()
outcome_vars = get_outcome_variables()

frequency_tables = {}

for pc, label_map in protected_classes.items():
    for dv in outcome_vars:
        # Compute and display the frequency table
        freq_table = compute_frequency_table(df, pc, dv)
        frequency_tables[(pc, dv)] = freq_table

        print(f"\n=== Frequency table: {pc} vs {dv} ===")
        display(freq_table)

        # Export to CSV for inclusion in the final report
        export_frequency_table(freq_table, pc, dv)

        # Generate and save bar plot of positive outcome rate
        plot_positive_rate_bar(df, pc, dv, label_map, save=True)
```

```
=== Frequency table: gender_bin vs dv_promoted ===

dv_promoted      0     1  total  positive_rate
gender_bin
```

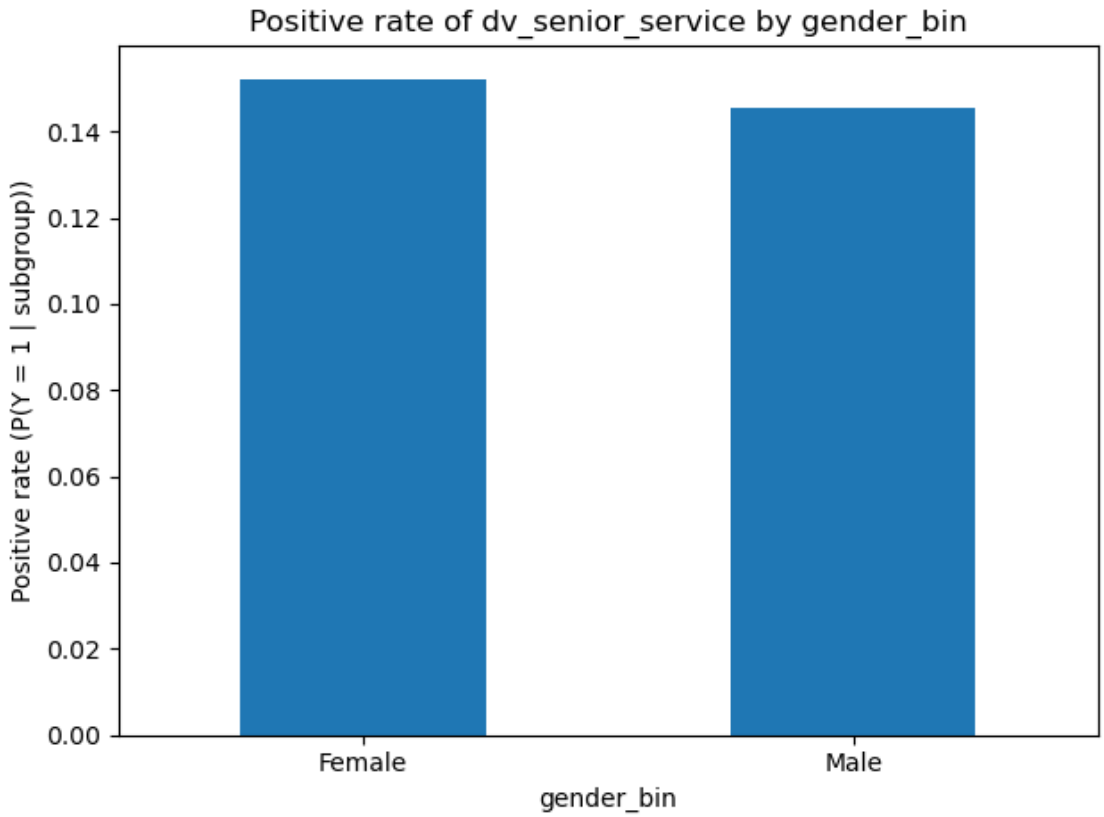
```
0              14845   1467   16312        0.089934
1              35295   3201   38496        0.083151
```



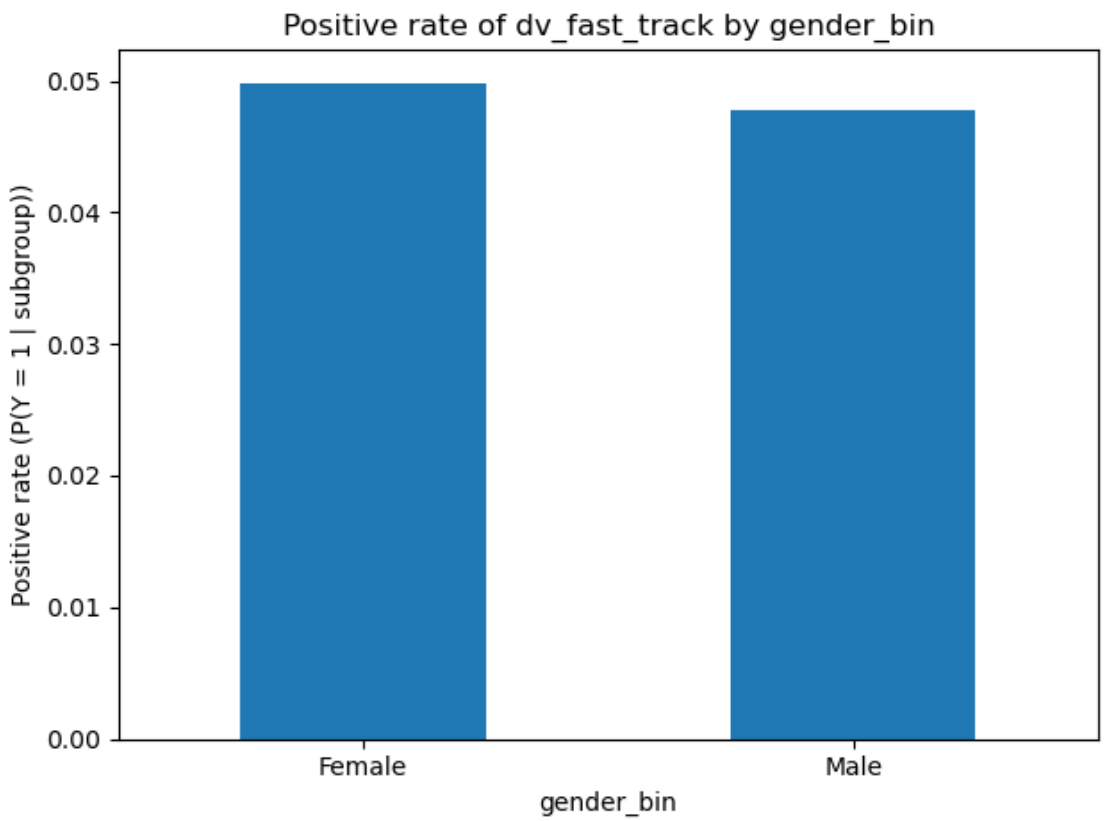

```
=== Frequency table: gender_bin vs dv_high_rating ===

dv_high_rating      0      1  total  positive_rate
gender_bin
0                9673   6639  16312       0.407001
1               23517  14979  38496       0.389105
```



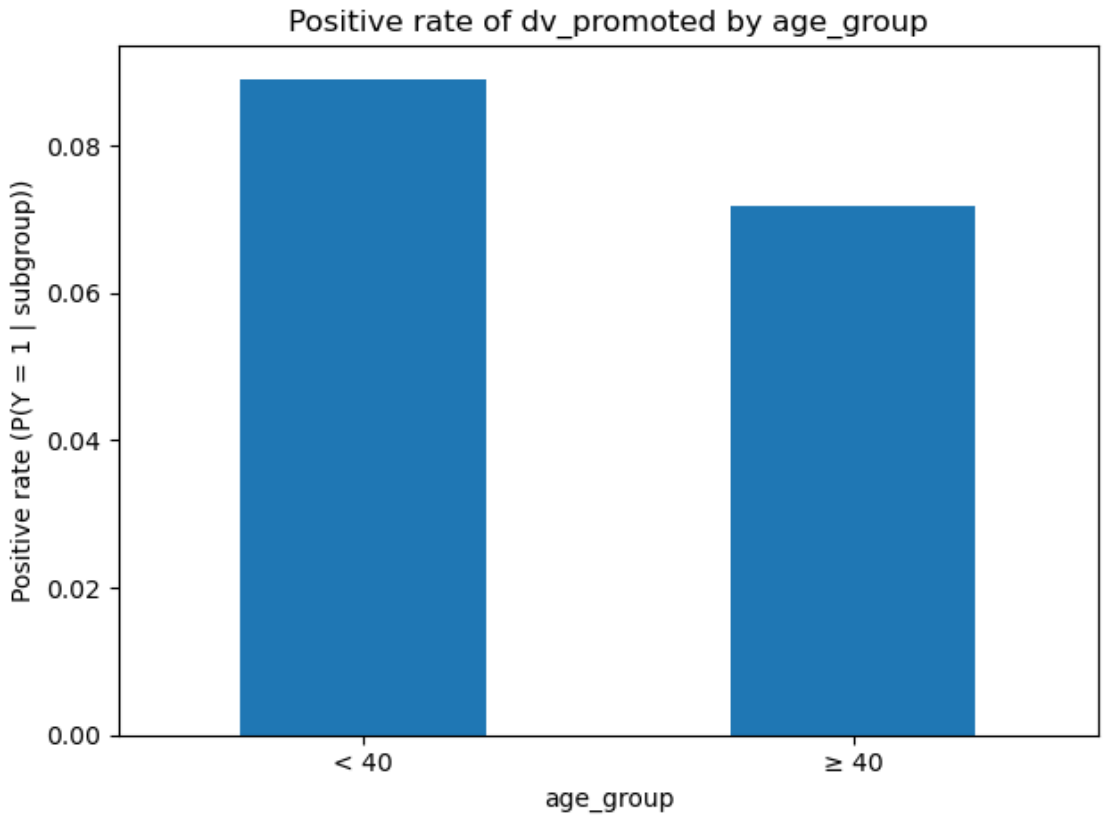

```
=== Frequency table: gender_bin vs dv_senior_service ===

dv_senior_service      0     1  total  positive_rate
gender_bin
0                  13829  2483  16312       0.152219
1                  32900  5596  38496       0.145366
```



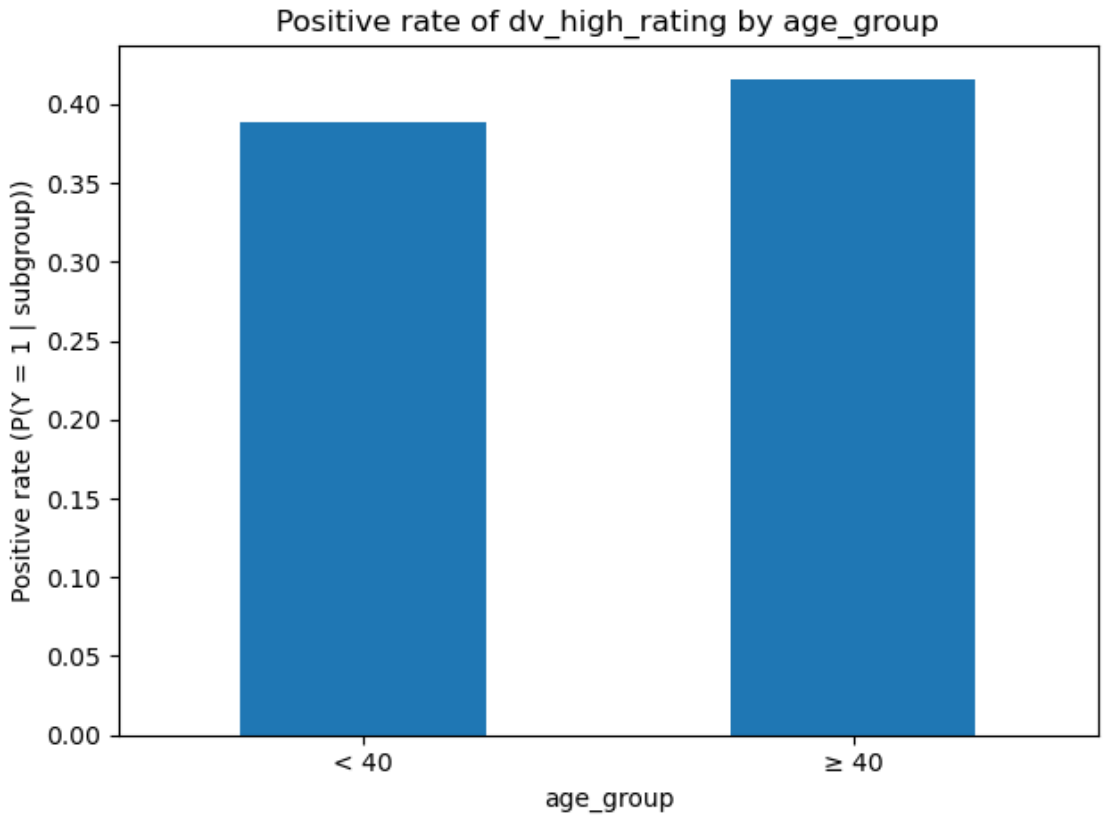

```
=== Frequency table: gender_bin vs dv_fast_track ===

dv_fast_track      0     1  total  positive_rate
gender_bin
0              15499   813  16312       0.049841
1              36655  1841  38496       0.047823
```



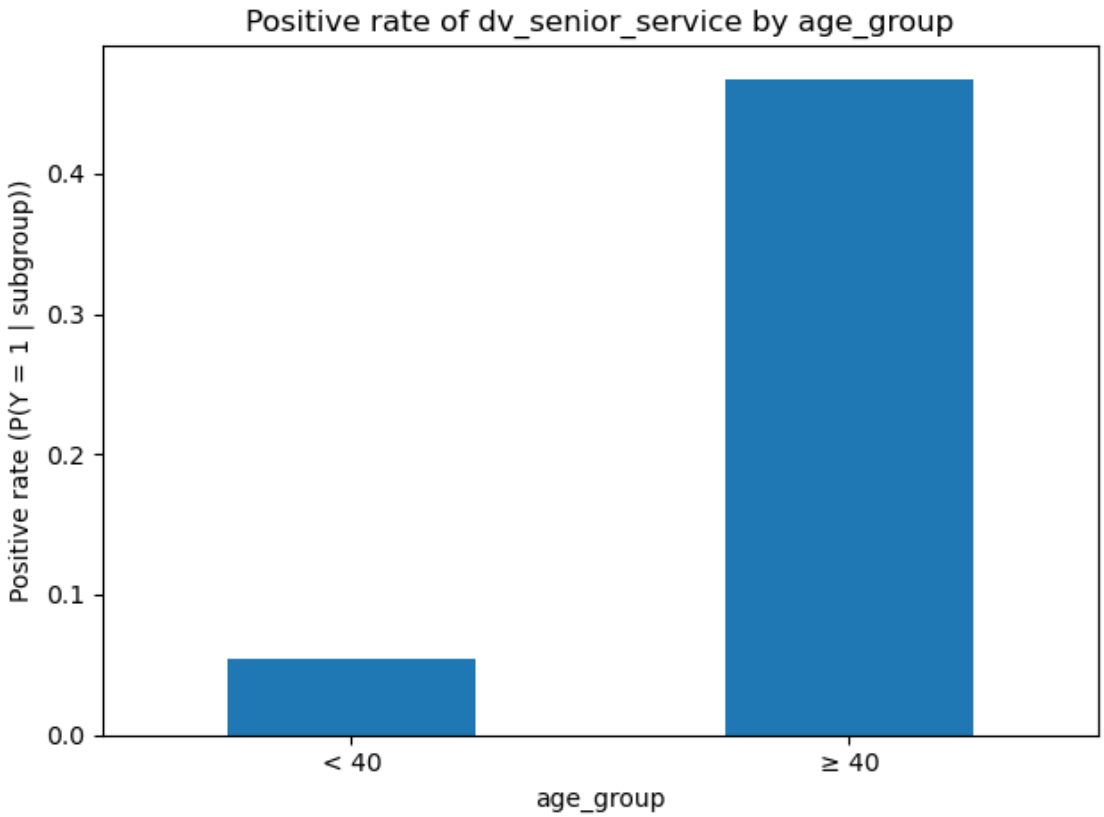

```
=== Frequency table: age_group vs dv_promoted ===

dv_promoted      0     1  total  positive_rate
age_group
0            38701  3782  42483       0.089024
1            11439   886  12325       0.071886
```



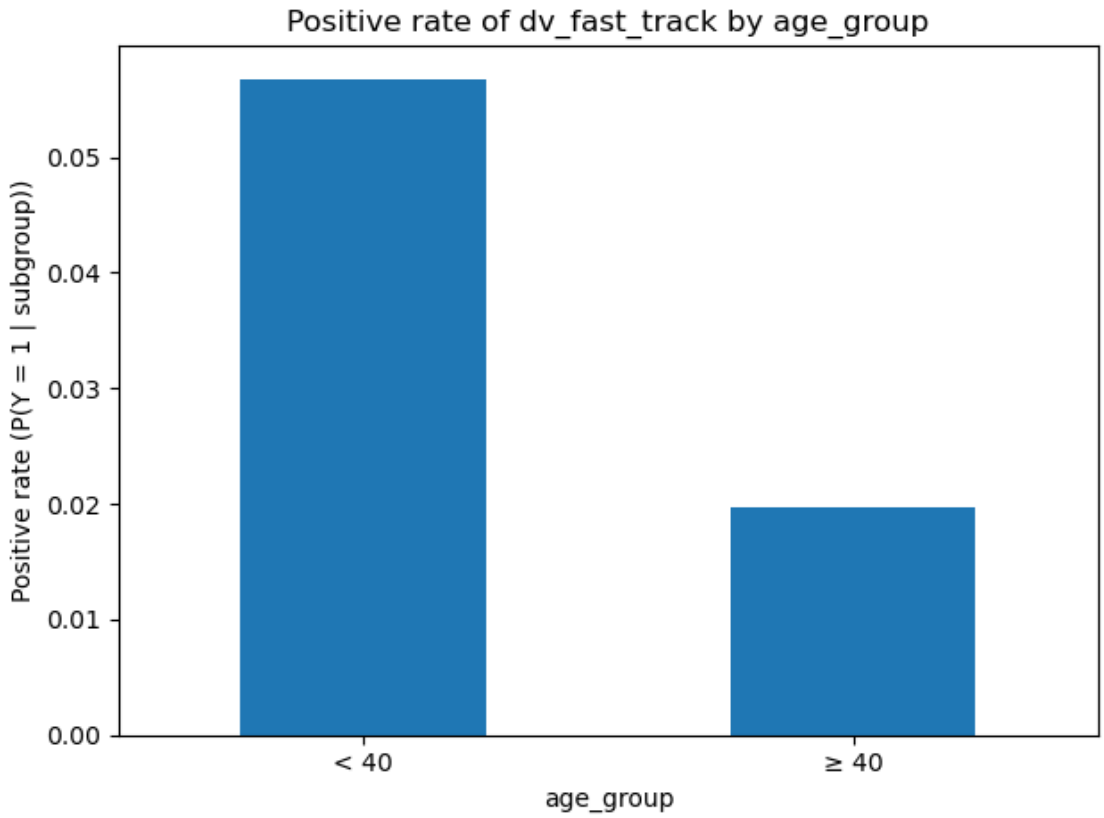

```
=== Frequency table: age_group vs dv_high_rating ===

dv_high_rating      0      1  total  positive_rate
age_group
0               25991  16492  42483       0.388202
1                7199   5126  12325       0.415903
```




```
=== Frequency table: age_group vs dv_senior_service ===

dv_senior_service      0     1  total  positive_rate
age_group
0                  40165  2318  42483       0.054563
1                   6564  5761  12325       0.467424
```




```
=== Frequency table: age_group vs dv_fast_track ===

dv_fast_track      0     1  total  positive_rate
age_group
0               40072  2411  42483       0.056752
1               12082   243  12325       0.019716
```



1.0.5 Step 3: Fairness Metrics and Baseline Assessment of Outcome Disparities

The objective of Step 3 is to formally quantify group-level disparities across all protected classes and outcome variables using established fairness metrics. This step provides the baseline fairness evaluation against which all subsequent modeling (Step 4) and mitigation (Step 5) results will be compared. By computing metrics directly on the **observed labels**, rather than model predictions, we characterize structural inequities present in the dataset itself before any algorithmic influence is introduced.

Two fairness metrics are used in accordance with project requirements:

1. **Statistical Parity Difference (SPD)**
   Measures the difference in the rate of favorable outcomes between the unprivileged and privileged groups:

   SPD = [P(Yhat = 1 | A = a_unpriv)] − [P(Yhat = 1 | A = a_priv)]

   Negative values indicate that the unprivileged group receives favorable outcomes less frequently.

2. **Equal Opportunity Difference (EOD)**
   Measures the difference in **true positive rates** between the unprivileged and privileged groups:

EOD = [TPR_(a_unpriv)] − [TPR_(a_priv)]

This metric evaluates whether qualified individuals in different groups have equal likelihood of receiving a positive classification.

These metrics are computed for each combination of: - The two legally recognized protected classes (`gender_bin`, `age_group`), and
- The four outcome variables (`dv_promoted`, `dv_high_rating`, `dv_senior_service`, `dv_fast_track`).

Computing SPD and EOD on the **ground-truth labels** provides critical insight into existing disparities prior to any modeling. This baseline will allow us to interpret whether predictive models amplify, reduce, or preserve these disparities. The functions defined below implement SPD and EOD in a reusable format, enabling consistent application across both label-based and model-based fairness evaluations in later sections.

[ ]:
```
# Step 3: Fairness metrics and baseline label-level disparities

import numpy as np
import pandas as pd


# ---------------------------------------------
# Fairness Metric Definitions
# ---------------------------------------------

def SPD(y_true, y_pred, group, unpriv=0, priv=1):
    """
    Statistical Parity Difference (SPD)
    -----------------------------------
    Computes the difference in the rate of favorable predictions between
    the unprivileged and privileged groups:

        SPD = P(ŷ = 1 | A = a_unpriv) - P(ŷ = 1 | A = a_priv)

    Parameters
    ----------
    y_true : array-like
        Ground truth labels (not used directly for SPD, included for API␣
 ↪consistency).
    y_pred : array-like
        Predicted labels (binary: 0/1).
    group : array-like
        Protected attribute values (binary: unprivileged/privileged).
    unpriv : int or float
        Value indicating the unprivileged group (default = 0).
    priv : int or float
        Value indicating the privileged group (default = 1).
```

```
    Returns
    -------
    float
        Statistical Parity Difference (SPD). Negative values indicate
        that the unprivileged group receives favorable outcomes less frequently.
    """
    y_pred = np.asarray(y_pred)
    group  = np.asarray(group)

    mask_unpriv = (group == unpriv)
    mask_priv   = (group == priv)

    # Avoid divide-by-zero errors
    if mask_unpriv.sum() == 0 or mask_priv.sum() == 0:
        return np.nan

    rate_unpriv = (y_pred[mask_unpriv] == 1).mean()
    rate_priv   = (y_pred[mask_priv] == 1).mean()

    return rate_unpriv - rate_priv



def EOD(y_true, y_pred, group, unpriv=0, priv=1):
    """
    Equal Opportunity Difference (EOD)
    ----------------------------------
    Computes the difference in true positive rates between the unprivileged
    and privileged groups:

        EOD = TPR_unprivileged - TPR_privileged

    Parameters
    ----------
    y_true : array-like
        Ground truth labels (binary: 0/1).
    y_pred : array-like
        Predicted labels (binary: 0/1).
    group : array-like
        Protected attribute values.
    unpriv : int or float
        Value representing the unprivileged group.
    priv : int or float
        Value representing the privileged group.

    Returns
    -------
```

```
    float
        Equal Opportunity Difference (EOD). Negative values indicate
        that qualified unprivileged individuals achieve favorable outcomes less␣
↪often.
    """
    y_true = np.asarray(y_true)
    y_pred = np.asarray(y_pred)
    group  = np.asarray(group)

    mask_unpriv = (group == unpriv)
    mask_priv   = (group == priv)

    # Select rows where true label is positive
    true_pos_unpriv = (y_true[mask_unpriv] == 1)
    true_pos_priv   = (y_true[mask_priv] == 1)

    if true_pos_unpriv.sum() == 0 or true_pos_priv.sum() == 0:
        return np.nan

    tpr_unpriv = (y_pred[mask_unpriv][true_pos_unpriv] == 1).mean()
    tpr_priv   = (y_pred[mask_priv][true_pos_priv] == 1).mean()

    return tpr_unpriv - tpr_priv



# ---------------------------------------------------------------
# Compute Baseline Fairness Using Ground Truth Labels
# ---------------------------------------------------------------

def compute_label_fairness(df, outcome_vars, protected_classes,␣
↪unpriv_priv_map):
    """
    Computes SPD and EOD using the ground-truth labels only.
    This provides the baseline structural disparity of the dataset.

    Parameters
    ----------
    df : pandas.DataFrame
        The full dataset with protected class columns and outcome variables.
    outcome_vars : list of str
        Dependent variables (e.g., ['dv_promoted', ...]).
    protected_classes : list of str
        Protected class attribute column names (e.g., ['gender_bin',␣
↪'age_group']).
    unpriv_priv_map : dict
        Maps each protected class to (unpriv, priv).
```

```
    Returns
    -------
    pandas.DataFrame
        Table of SPD and EOD for each PC × DV pair.
    """
    records = []

    for dv in outcome_vars:
        y = df[dv].values

        for pc in protected_classes:
            group = df[pc].values
            unpriv, priv = unpriv_priv_map[pc]

            spd_val = SPD(y_true=y, y_pred=y, group=group, unpriv=unpriv,␣
 ↪priv=priv)
            eod_val = EOD(y_true=y, y_pred=y, group=group, unpriv=unpriv,␣
 ↪priv=priv)

            records.append({
                'protected_class': pc,
                'outcome_variable': dv,
                'unprivileged_value': unpriv,
                'privileged_value': priv,
                'SPD_label_level': spd_val,
                'EOD_label_level': eod_val
            })

    return pd.DataFrame(records)

# ---------------------------------------------------------------
# Execute Label-Level Fairness Evaluation
# ---------------------------------------------------------------

outcome_vars = ['dv_promoted', 'dv_high_rating', 'dv_senior_service',␣
 ↪'dv_fast_track']
protected_classes = ['gender_bin', 'age_group']

# Defines which numeric value is unpriv and which is priv
unpriv_priv_map = {
    'gender_bin': (0, 1),      # Female (0) = unpriv, Male (1) = priv
    'age_group':  (1, 0)       # Age   40 = unpriv, Age < 40 = priv
}

label_fairness_df = compute_label_fairness(
    df,
```

```
    outcome_vars=outcome_vars,
    protected_classes=protected_classes,
    unpriv_priv_map=unpriv_priv_map
)

label_fairness_df.to_csv("artifacts/label_fairness_metrics.csv", index=False)

# Display results
label_fairness_df
```

[ ]:

```
  protected_class   outcome_variable  unprivileged_value  privileged_value  \
0      gender_bin        dv_promoted                   0                 1
1       age_group        dv_promoted                   1                 0
2      gender_bin     dv_high_rating                   0                 1
3       age_group     dv_high_rating                   1                 0
4      gender_bin  dv_senior_service                   0                 1
5       age_group  dv_senior_service                   1                 0
6      gender_bin      dv_fast_track                   0                 1
7       age_group      dv_fast_track                   1                 0

   SPD_label_level  EOD_label_level
0         0.006782              0.0
1        -0.017137              0.0
2         0.017896              0.0
3         0.027700              0.0
4         0.006853              0.0
5         0.412861              0.0
6         0.002017              0.0
7        -0.037036              0.0
```

1.0.6 Step 4: Mitigating Bias with Classification Models

The goal of Step 4 is to evaluate whether the **pre-processing bias mitigation** applied in Step 3 (reweighing based on one dependent variable) is effective once a predictive model is trained. In this step, I train and evaluate classifiers on both the **original** and **transformed** datasets and compare fairness metrics across the two settings.

Consistent with the project instructions, I focus on a **single primary outcome variable** as the prediction target:

- **Primary label for classification:** `dv_promoted` (promotion outcome)

All other engineered variables (excluding identifiers and outcome variables) are treated as potential predictors. Protected class attributes (`gender_bin`, `age_group`) are **not used as input features** for the classifier but are retained in the dataset to compute fairness metrics on the resulting predictions.

Step 4 is organized into two parallel pipelines:

1. **Original Dataset Pipeline**

1. Randomly split the original dataset into training and testing sets (70/30 split with a fixed random seed for reproducibility).

2. Train a baseline classifier (Logistic Regression) to predict `dv_promoted` using the selected feature set.

3. On the held-out test set, compute:
   - Standard performance metrics (accuracy, ROC–AUC, confusion matrix), and

   - Fairness metrics (SPD and EOD) for both protected classes:
     - `gender_bin` (Female = unprivileged, Male = privileged)

     - `age_group` (Age 40 = unprivileged, Age < 40 = privileged)

2. **Transformed Dataset Pipeline**
   1. Use the **bias-mitigated dataset** produced in Step 3 (reweighted or otherwise transformed as a function of `dv_promoted`).
   2. Randomly split the transformed dataset into training and testing sets using the same split ratio and random seed strategy.

   3. Train the same classifier type (Logistic Regression with identical hyperparameters) on the transformed training data.

   4. On the transformed test set, compute the same performance and fairness metrics (SPD and EOD) for the same protected classes.

For each protected class and fairness metric, I then construct a **summary table** that compares:

- Label-level fairness on the original dataset (from Step 3),

- Fairness on model predictions from the original dataset, and

- Fairness on model predictions from the transformed dataset.

This table allows me to determine whether the pre-processing bias mitigation produced a **positive change, negative change, or no change** in SPD and EOD, both at the data level and after training the classifier, as required by the project specification.

[55]:
```
from sklearn.impute import SimpleImputer

def prepare_features_and_labels(df, label_col, protected_cols,␣
 ↪exclude_cols=None):
    """
    Prepares feature matrix X, label vector y, and protected class arrays
    for model training and evaluation. Handles missing values by imputing
    numeric features with the median and categorical features with the
    most frequent category.

    Parameters
    ----------
```

```
    df : pandas.DataFrame
        Input dataset with engineered outcome and protected class columns.
    label_col : str
        Name of the target variable (e.g., 'dv_promoted').
    protected_cols : list of str
        Columns corresponding to protected attributes (e.g., ['gender_bin',␣
↪'age_group']).
    exclude_cols : list of str, optional
        Columns to exclude from the feature set (IDs, labels, protected␣
↪attributes).

    Returns
    -------
    X : pandas.DataFrame
        One-hot encoded, imputed feature matrix.
    y : pandas.Series
        Target labels.
    protected_data : dict
        Mapping from protected class column name to its Series.
    """
    if exclude_cols is None:
        exclude_cols = []

    # Label
    y = df[label_col].astype(int)

    # Protected attributes (not used as features)
    protected_data = {pc: df[pc] for pc in protected_cols}

    # Build base feature set by excluding label, protected attributes, and␣
↪other specified cols
    cols_to_exclude = set(exclude_cols + [label_col] + protected_cols)
    feature_cols = [c for c in df.columns if c not in cols_to_exclude]

    X_raw = df[feature_cols].copy()

    # Split into numeric and non-numeric for different imputation strategies
    num_cols = X_raw.select_dtypes(include=['number']).columns.tolist()
    cat_cols = [c for c in X_raw.columns if c not in num_cols]

    # Impute numeric features with median
    if num_cols:
        num_imputer = SimpleImputer(strategy='median')
        X_raw[num_cols] = num_imputer.fit_transform(X_raw[num_cols])

    # Impute categorical features with most frequent category
    if cat_cols:
```

```
        cat_imputer = SimpleImputer(strategy='most_frequent')
        X_raw[cat_cols] = cat_imputer.fit_transform(X_raw[cat_cols])

    # One-hot encode categorical features
    X = pd.get_dummies(X_raw, columns=cat_cols, drop_first=True)

    return X, y, protected_data
```

[56]:
```
# ------------------------------------------------------------
# 1. Build features, labels, and protected attributes for dv_promoted
# ------------------------------------------------------------

protected_cols = ['gender_bin', 'age_group']

# Columns you do NOT want as features (IDs, engineered outcomes, weights, etc.)
exclude_cols = [
    'dv_promoted',
    'dv_high_rating',
    'dv_senior_service',
    'dv_fast_track',
    'rw_weight_gender_promo',  # if present
    'age_bucket'               # if present
]

X, y, protected_data = prepare_features_and_labels(
    df=df,
    label_col='dv_promoted',
    protected_cols=protected_cols,
    exclude_cols=exclude_cols
)
```

[57]:
```
# ------------------------------------------------------------
# 2. Train/test split and scaling
# ------------------------------------------------------------

RANDOM_STATE = 42

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.30,
    random_state=RANDOM_STATE,
    stratify=y
)

# Align protected attributes with the split indices
gender_train = protected_data['gender_bin'].loc[X_train.index]
gender_test  = protected_data['gender_bin'].loc[X_test.index]
```

```
age_train    = protected_data['age_group'].loc[X_train.index]
age_test     = protected_data['age_group'].loc[X_test.index]

# Scale features (without centering to preserve sparse structure)
scaler = StandardScaler(with_mean=False)
X_train_sc = scaler.fit_transform(X_train)
X_test_sc  = scaler.transform(X_test)
```

[58]:
```
# ------------------------------------------------------------
# Rebuild features, labels, and protected attributes cleanly
# ------------------------------------------------------------

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

protected_cols = ['gender_bin', 'age_group']

exclude_cols = [
    'dv_promoted',
    'dv_high_rating',
    'dv_senior_service',
    'dv_fast_track',
    'rw_weight_gender_promo',
    'age_bucket',
    'is_promoted'
]

# Use your existing helper (no weight_col argument)
X, y, protected_attrs = prepare_features_and_labels(
    df=df,
    label_col='dv_promoted',
    protected_cols=protected_cols,
    exclude_cols=exclude_cols
)

# Train/test split ONLY on X and y
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    stratify=y,
    random_state=RANDOM_STATE
)

# Align protected attributes to the split using the indices
gender_train = protected_attrs['gender_bin'].loc[X_train.index]
gender_test  = protected_attrs['gender_bin'].loc[X_test.index]
```

```
age_train    = protected_attrs['age_group'].loc[X_train.index]
age_test     = protected_attrs['age_group'].loc[X_test.index]

# Scale cleanly
scaler = StandardScaler(with_mean=False)
X_train_sc = scaler.fit_transform(X_train)
X_test_sc  = scaler.transform(X_test)
```

[59]:
```
clf_lr = train_logistic_regression(X_train_sc, y_train)

y_pred_lr  = clf_lr.predict(X_test_sc)
y_proba_lr = clf_lr.predict_proba(X_test_sc)[:, 1]

acc_lr = accuracy_score(y_test, y_pred_lr)
auc_lr = roc_auc_score(y_test, y_proba_lr)

print(f"Logistic Regression - Accuracy: {acc_lr:.4f}, ROC-AUC: {auc_lr:.4f}")
print("Confusion matrix:\n", confusion_matrix(y_test, y_pred_lr))
```

```
Logistic Regression - Accuracy: 0.9388, ROC-AUC: 0.7913
Confusion matrix:
 [[15025    18]
 [  988   412]]
```

[60]:
```
# Unprivileged/privileged mapping
unpriv_priv_map = {
    'gender_bin': (0, 1),   # Female = unprivileged, Male = privileged
    'age_group':  (1, 0)    # Age   40 = unprivileged, Age < 40 = privileged
}

def evaluate_fairness_for_model(y_true, y_pred, protected_series, unpriv, priv,␣
 ↪pc_name):
    g = protected_series.values
    return {
        'protected_class': pc_name,
        'unprivileged_value': unpriv,
        'privileged_value': priv,
        'SPD_pred': SPD(y_true=y_true, y_pred=y_pred, group=g, unpriv=unpriv,␣
 ↪priv=priv),
        'EOD_pred': EOD(y_true=y_true, y_pred=y_pred, group=g, unpriv=unpriv,␣
 ↪priv=priv),
    }

fairness_results = []

# Gender
fairness_results.append(
```

```
    evaluate_fairness_for_model(
        y_true=y_test.values,
        y_pred=y_pred_lr,
        protected_series=gender_test,
        unpriv=unpriv_priv_map['gender_bin'][0],
        priv=unpriv_priv_map['gender_bin'][1],
        pc_name='gender_bin'
    )
)

# Age
fairness_results.append(
    evaluate_fairness_for_model(
        y_true=y_test.values,
        y_pred=y_pred_lr,
        protected_series=age_test,
        unpriv=unpriv_priv_map['age_group'][0],
        priv=unpriv_priv_map['age_group'][1],
        pc_name='age_group'
    )
)


fairness_lr_df = pd.DataFrame(fairness_results)
fairness_lr_df
```

[60]:

	protected_class	unprivileged_value	privileged_value	SPD_pred	EOD_pred
0	gender_bin	0	1	0.006574	0.022134
1	age_group	1	0	-0.006726	-0.037500

1.0.7 Step 4 (continued): Mitigated Model via Reweighing

In this section, I apply a pre-processing bias mitigation technique to the primary outcome variable `dv_promoted`. Specifically, I use a *reweighing* strategy based on the protected attribute `gender_bin`, which adjusts the effective sample weights for combinations of (A, Y), where A is gender and Y is the promotion label.

The reweighted training data are then used to fit a Logistic Regression classifier with the same feature set and hyperparameters as the baseline model. I evaluate:

- Standard performance metrics (accuracy, ROC–AUC), and
- Fairness metrics (SPD and EOD) for `gender_bin` and `age_group`

on the held-out test set. These results are compared to the baseline model to assess whether reweighing improves group fairness and what impact it has on predictive performance.

The Kamiran & Calders reweighing method computes weights for each protected-class and label pair using:

w(a,y) = P(A=a) * P(Y=y) / P(A=a, Y=y)

This formula explicitly corrects imbalances between privileged and unprivileged groups at the label level by adjusting sample weights prior to training.

[61]:
```
# Step 4 (continued): Reweighing for dv_promoted using gender_bin

def compute_reweighing_weights(df, label_col, group_col):
    """
    Computes Kamiran & Calders-style reweighing weights for each combination
    of protected attribute value and label.

    Weight for each (A=a, Y=y) is:
        w(a,y) = P(A=a) * P(Y=y) / P(A=a, Y=y)

    Parameters
    ----------
    df : pandas.DataFrame
        Data containing the label and protected attribute.
    label_col : str
        Name of the label column (e.g., 'dv_promoted').
    group_col : str
        Name of the protected attribute column (e.g., 'gender_bin').

    Returns
    -------
    pandas.Series
        Sample weights aligned with df.index.
    """
    # Joint counts
    joint = df.groupby([group_col, label_col]).size().rename("count").
 ↪reset_index()

    N = len(df)

    # Marginals
    p_a = joint.groupby(group_col)['count'].sum() / N
    p_y = joint.groupby(label_col)['count'].sum() / N

    # Joint probability
    joint['p_ay'] = joint['count'] / N

    # Attach marginals
    joint['p_a'] = joint[group_col].map(p_a)
    joint['p_y'] = joint[label_col].map(p_y)

    # Compute weights
    joint['weight'] = (joint['p_a'] * joint['p_y']) / joint['p_ay']
```

```
    # Map back to each row in df
    w = df[[group_col, label_col]].merge(
        joint[[group_col, label_col, 'weight']],
        on=[group_col, label_col],
        how='left'
    )['weight']

    return w


# Compute weights on the full dataframe for dv_promoted × gender_bin
df['rw_weight_gender_promo'] = compute_reweighing_weights(
    df=df,
    label_col='dv_promoted',
    group_col='gender_bin'
)

df['rw_weight_gender_promo'].describe()
```

[61]:
```
count    54808.000000
mean         1.000000
std          0.010957
min          0.947031
25%          0.997798
50%          0.997798
75%          1.005235
max          1.024276
Name: rw_weight_gender_promo, dtype: float64
```

[66]:
```
# --- Reweighed model for dv_promoted using gender_bin ---

label_col = 'dv_promoted'
protected_cols = ['gender_bin', 'age_group']

# Rebuild features/labels
X_rw, y_rw, protected_data_rw = prepare_features_and_labels(
    df=df,
    label_col=label_col,
    protected_cols=protected_cols,
    exclude_cols=exclude_cols
)

# Train/test split (same RANDOM_STATE and ratio for comparability)
X_train_rw, X_test_rw, y_train_rw, y_test_rw = train_test_split(
    X_rw, y_rw,
    test_size=0.30,
```

```
    random_state=RANDOM_STATE,
    stratify=y_rw
)

# Protected attributes aligned with the split
gender_train_rw = protected_data_rw['gender_bin'].loc[X_train_rw.index]
gender_test_rw  = protected_data_rw['gender_bin'].loc[X_test_rw.index]
age_train_rw    = protected_data_rw['age_group'].loc[X_train_rw.index]
age_test_rw     = protected_data_rw['age_group'].loc[X_test_rw.index]

# Training weights (reweighing) aligned with the training rows only
w_train_rw = df.loc[X_train_rw.index, 'rw_weight_gender_promo']

# Scale features (fit on train, transform on both)
X_train_rw_sc = scaler.fit_transform(X_train_rw)
X_test_rw_sc  = scaler.transform(X_test_rw)

# Train Logistic Regression with sample weights
clf_lr_rw = LogisticRegression(max_iter=1000, solver='lbfgs')
clf_lr_rw.fit(X_train_rw_sc, y_train_rw, sample_weight=w_train_rw)

# Evaluate performance on the reweighed model
y_pred_rw = clf_lr_rw.predict(X_test_rw_sc)
y_proba_rw = clf_lr_rw.predict_proba(X_test_rw_sc)[:, 1]

acc_rw = accuracy_score(y_test_rw, y_pred_rw)
auc_rw = roc_auc_score(y_test_rw, y_proba_rw)

print(f"Reweighed LR - Accuracy: {acc_rw:.4f}, ROC-AUC: {auc_rw:.4f}")
print("Confusion matrix:\n", confusion_matrix(y_test_rw, y_pred_rw))
```

```
Reweighed LR - Accuracy: 0.9386, ROC-AUC: 0.7906
Confusion matrix:
 [[15023    20]
 [  989   411]]
```

[63]:
```
fairness_results_rw = []

# Gender fairness (reweighed model)
fairness_results_rw.append(
    evaluate_fairness_for_model(
        y_true=y_test_rw.values,
        y_pred=y_pred_rw,
        protected_series=gender_test_rw,
        unpriv=unpriv_priv_map['gender_bin'][0],
        priv=unpriv_priv_map['gender_bin'][1],
        pc_name='gender_bin'
```

```
    )
)

# Age fairness (reweighed model)
fairness_results_rw.append(
    evaluate_fairness_for_model(
        y_true=y_test_rw.values,
        y_pred=y_pred_rw,
        protected_series=age_test_rw,
        unpriv=unpriv_priv_map['age_group'][0],
        priv=unpriv_priv_map['age_group'][1],
        pc_name='age_group'
    )
)

fairness_lr_rw_df = pd.DataFrame(fairness_results_rw)
fairness_lr_rw_df['model'] = 'LogisticRegression_reweighed'
fairness_lr_rw_df
```

[63]:

```
  protected_class  unprivileged_value  privileged_value  SPD_pred  EOD_pred  \
0      gender_bin                   0                 1  0.005897  0.016689
1       age_group                   1                 0 -0.006805 -0.036607

                          model
0  LogisticRegression_reweighed
1  LogisticRegression_reweighed
```

[64]:

```
import os

os.makedirs("artifacts/model_results", exist_ok=True)

# Fairness metrics: original LR model on dv_promoted
fairness_lr_df.to_csv(
    "artifacts/model_results/fairness_lr_original.csv",
    index=False
)

# Fairness metrics: reweighed LR model on dv_promoted
fairness_lr_rw_df.to_csv(
    "artifacts/model_results/fairness_lr_reweighed.csv",
    index=False
)

# Performance summary for both models
from sklearn.metrics import confusion_matrix

tn, fp, fn, tp = confusion_matrix(y_test, y_pred_lr).ravel()
```

```
tn_rw, fp_rw, fn_rw, tp_rw = confusion_matrix(y_test_rw, y_pred_rw).ravel()

perf_summary = pd.DataFrame([
    {
        "model": "LogisticRegression_original",
        "accuracy": acc_lr,
        "roc_auc": auc_lr,
        "tn": tn, "fp": fp, "fn": fn, "tp": tp,
    },
    {
        "model": "LogisticRegression_reweighed",
        "accuracy": acc_rw,
        "roc_auc": auc_rw,
        "tn": tn_rw, "fp": fp_rw, "fn": fn_rw, "tp": tp_rw,
    }
])

perf_summary.to_csv(
    "artifacts/model_results/performance_lr_models.csv",
    index=False
)
```

1.0.8 Step 5: Fairness Comparison Across Label-Level, Original Model, and Mitigated Model

In this step, I consolidate all fairness results for the primary dependent variable `dv_promoted` into a single comparison table. This table includes:

- Label-level fairness metrics** (from Step 3), which measure pre-existing structural disparities in the raw dataset.
- Original logistic regression model metrics** (from Step 4), which measure how the baseline classifier propagates or amplifies those disparities.
- Reweighed model metrics** (from Step 4 mitigation), which measure whether the pre-processing reweighing technique improved fairness for the protected class targeted during mitigation.

The comparison focuses on two legally recognized protected classes:

- `gender_bin` — Female (0, unprivileged) vs. Male (1, privileged)

- `age_group` — Age 40 (1, unprivileged) vs. Age < 40 (0, privileged)

For each protected class, I compare:

- **Statistical Parity Difference (SPD):**
  Difference in predicted positive outcomes between the unprivileged and privileged groups.

- **Equal Opportunity Difference (EOD):**
  Difference in true positive rates between the unprivileged and privileged groups.

This table allows me to directly evaluate whether reweighing:

1. Moved SPD and EOD **closer to zero** (improved fairness),

2. Moved them **further from zero** (reduced fairness), or

3. Produced **mixed results**, improving one metric but not the other.

This analysis forms the basis for the final discussion and conclusion about the effectiveness and tradeoffs of the mitigation technique applied in Step 4.

[65]:
```
# ---- Step 5: Fairness comparison table for dv_promoted ----

# 1. Prepare label-level fairness data for dv_promoted
label_promo = label_fairness_df.copy()

# Rename columns to standard names expected in comparison
label_promo = label_promo.rename(columns={
    'SPD_label_level': 'SPD_label',
    'EOD_label_level': 'EOD_label'
})

# Filter only dv_promoted
label_promo = label_promo[label_promo['outcome_variable'] == 'dv_promoted'].
 ↪copy()

# 2. Prepare original model fairness
fairness_lr_df_promo = fairness_lr_df.copy()
fairness_lr_df_promo['outcome_variable'] = 'dv_promoted'
fairness_lr_df_promo = fairness_lr_df_promo.rename(columns={
    'SPD_pred': 'SPD_pred_original',
    'EOD_pred': 'EOD_pred_original'
})

# 3. Prepare reweighed model fairness
fairness_lr_rw_df_promo = fairness_lr_rw_df.copy()
fairness_lr_rw_df_promo['outcome_variable'] = 'dv_promoted'
fairness_lr_rw_df_promo = fairness_lr_rw_df_promo.rename(columns={
    'SPD_pred': 'SPD_pred_reweighed',
    'EOD_pred': 'EOD_pred_reweighed'
})

# 4. Merge label-level + original model
comparison = label_promo.merge(
    fairness_lr_df_promo[['protected_class', 'outcome_variable',
                          'SPD_pred_original', 'EOD_pred_original']],
    on=['protected_class', 'outcome_variable'],
    how='left'
)
```

```python
# 5. Merge in reweighed model results
comparison = comparison.merge(
    fairness_lr_rw_df_promo[['protected_class', 'outcome_variable',
                             'SPD_pred_reweighed', 'EOD_pred_reweighed']],
    on=['protected_class', 'outcome_variable'],
    how='left'
)

# 6. Final clean ordering
comparison = comparison[[
    'protected_class', 'outcome_variable',
    'SPD_label', 'EOD_label',
    'SPD_pred_original', 'EOD_pred_original',
    'SPD_pred_reweighed', 'EOD_pred_reweighed'
]]

comparison

comparison.to_csv("artifacts/model_results/fairness_comparison_dv_promoted.
 ↪csv", index=False)
```